

Ideomotor Principle as a Human-Robot Communication Method during Collaborative Work

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ABSTRACT

Body language is an essential component of communication. The amount of unspoken information it transmits during interpersonal interactions is an invaluable complement to simple speech and makes the process smoother and more sustainable. On the contrary, existing approaches to human-machine collaboration and communication are not as intuitive. This is an issue that needs to be addressed if we aim to continue using artificial intelligence and machines to increase our cognitive or even physical capabilities.

In this study, we analyse the potential of an intuitive communication method between biological and artificial agents, based on machines understanding and learning the subtle unspoken and involuntary cues found in human motion during the interaction process. Our work was divided into two stages: the first, analysing whether a machine using these implicit cues would produce the same positive effect as when they are manifested in interpersonal communication; the second, evaluating whether a machine could identify the cues manifested in human motion and learn to associate them with the appropriate command intended from its user. Results showed promising results with improved work performance and reduced cognitive load on the user side when relying on the proposed method, hinting to the potential of more intuitive, human to human inspired, communication methods in human-machine interaction.

CCS CONCEPTS

• **Human-centered computing** → **Gestural input**; • **Networks** → **Programming interfaces**; • **Computer systems organization** → **Robotic control**.

KEYWORDS

datasets, neural networks, gaze detection, text tagging

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1 INTRODUCTION

With the progress of robotics, individuals are becoming used to the presence of artificial agents in their household and workplace environments, and to interact with them on a daily basis. Furthermore, the boom of research in the field of "Robot Autonomy" ([10]), underlined the enthusiasm for autonomous robots capable of task completion with the same dexterity level as an individual ([7, 12]). Robots becoming more and more intelligent, capable of a higher level of understanding, has created a need for a shift in communication methods and the way artificial agents and biological agents interact.

While traditional methods such as direct teaching or action coding ([13]) may still work for most of the current applications of robotics, they are highly context-dependant with a high task specific dependency, and hence become very limited when used in uncontrolled environments (common in the field of service robotics). This implies that the user has to focus his/her attention on controlling or supervising the actions executed by the robot instead of performing his/her own task. More recently, some advances have been made, using interpersonal communication as inspiration for Human-Robot Interaction models, for example, Sun et al. [11] developed a method for a robot to infer the appropriate behaviour (in a conversation setting) based on the user's apparent engagement. Nevertheless, such methods lead to the robot taking on a more passive role unless the user is fully engaged in the interaction, leaving very little space for the individual to focus on anything else than operating/interacting with the robot.

In the present projects, we analysed the potential of an "implicit" communication method between a robot and its user during cooperative work. In human communication, the existence of such subtle non-verbal cues, manifesting themselves regardless of the person's intention is a known phenomenon and well-studied in psychology and neurosciences ([1, 4, 6]). We use an approach to human machine interaction similar to the Clever Hans Effect [8], where the robot becomes capable of understanding the action expected of it by responding to the involuntary and unconscious cuing (translated into postural adjustments) that occur in human communication (referred to as ideomotor reactions). Previous HRI studies [9] have shown that a highly proactive robot (making strong assumptions and actively acting on them) was often negatively perceived by users. Hence, in this study we see the use of ideomotor cues as a way for the robot to confirm assumptions (instead of directly acting upon them), without requiring any additional effort from the user. Because most studies on human-machine interaction are often one sided, focusing only on how the "human" element can instruct the machine, we chose to analyse, not only if these cues were a viable

communication method for the user to operate the machine, but also for the machine to communicate information to its user.

2 RESEARCH APPROACH:

2.1 Step 1: Machine to human information transmission

For the first part of the analysis, data collection, Japanese rice cake making was selected as a collaborative activity satisfying the following two elements:

- (1) Two individuals working together
- (2) Each individual's actions depend on the response of the other

The rice cake making process consists in one person pounding the cake while the second kneads and turns over the dough requires perfect synchronisation for optimal work and safety. In case of an eventual change in the pace of one of the two parties, the other needs to be able to anticipate and adapt his own behaviour accordingly.

Using the Unity game engine, we built a method for measuring the potential implicit cues used during Japanese rice cake (mochi) making. Figure 1 is a representation of the designed environment. Movement of the pestle was measured and recorded by attaching HTC VIVE trackers (a motion tracking accessory, calculating its position based on infrared signals emitted by virtual reality base stations [3]) to its handle and head. VIVE trackers were also attached to each glove worn by the person kneading the dough. The mortar and dough were respectively represented by a 0.5m high stool with a seat of 0.4m, and a disk shaped sponge with a radius of 0.3m. Participants were asked to position themselves on opposite sides of the stool, facing each other. Task execution performance of all participant The experiment was conducted with 14 participants divided into two groups (between-subjects format). Each participant of one group performed the task with the seven participants of the other. pairs was analyzed qualitatively and qualitatively. Motion patterns of the participants consistently displaying high performance was thoroughly analysed and compared to those of participants with lower results.

The second half of this analysis was done in the virtual reality (VR) environment shown in Figure 2. The system requires a computer, a Head Mounted Display (HMD), and a pestle with VIVE trackers attached to it (identically to the ones mentioned in previous section). This time, the kneading of the dough is performed by a robot arm in the virtual reality environment (Figure 2). Here, the participant works in tandem with the robot arm on the making of the rice cake. The robot arm was programmed to perform similar motion cues as observed in the collaboration between two participants when it required a specific action from the participant. The goal was to determine if the implicit cues identified during the first half of the experiment (with two participants) could be mimicked by a robot and still produce the same effect (nonverbal communication/understanding) during a human-machine collaboration instance.

2.2 Step 2: Human to machine information transmission

During the second step of our study, we analyzed the ability of an artificial agent to autonomously recognise ideomotor cues as commands (manifestation of the desire of the user for the robot to perform a specific action).

A simple network (Figure 3 was designed for the robot to learn the implicit cues and associate them with intended commands. As can be seen on Figure 3, the network is composed of concatenated Convolutional Neural Networks (CNN, 5 layers) followed by two layers of Long Short-Term Memory (LSTM) Network. The "CuDNNLSTM" in the Keras library was used as an accelerator for the LSTM. User movement input data collected through the use of Inertial Measurement Unit (IMU) sensors and robot positional data (motor encoder data) are used as input data. The IMUs are used to collect user motion data from 4 different locations (attached to the individual's body). The sensor model used was TSND151 [2], with a data output of 10 dimensions: 3 axes of acceleration, 3 axes of angular velocity and posture (quaternion = 4 dimensions). For the robot arm, data input into the network consisted of the angle data on each axis of the arm (0° to 180°, 4 degrees of freedom). During collection of the training data, matching label for each movement of the user was collected by having them vocally express which action they wished for the robot arm to perform at that instant (in the designed experiment, 5 options were available: reach, grasp, release, return, wipe). To verify the capacity of the robot to learn and identify the implicit cues in real time, the experiments pertaining to this section were conducted using a real robot.

The tasks used for this phase of the experiment were designed to mimic daily chores that an individual may perform, while remaining relatively simple as to keep the number of commands and labels relatively low (4 different labels and a standby phase). 3 Tasks (Figure 4, Task1, Task 2, Task 3) were "periodic" with a constant label order, one task (Task 4) was "aperiodic" with the order of some labels randomly changing. While task 2 and task 4 may seem similar, in task 2, the user always performed the same motions in the same order, in task 4 motions corresponding to each label could be performed in any order.

3 RESULTS AND FINDINGS: IMPROVED WORK EFFICIENCY AND MULTI-TASKING

3.1 Machine mimicking ideomotor cues

Qualitative evaluation of the Human-Machine collaboration experiment was performed using a 7-grade Likert scale survey. Survey questions are shown in Table 1. As shown by results gathered in Figure 5, both questions received much better ratings when the artificial agent made use of the mimicked ideomotor cues. Work performance improvement was also reflected in the quantitative work evaluation results. As reflected in Figure 6, participants were able to correctly match a sudden work speed increase demand by the robot (in this example 1.66 seconds between two punches) when it used motion cues. On the contrary, when these cues were not used, participants tended to be surprised by the sudden behavioural change of the robot (change in working pace) and would slow down

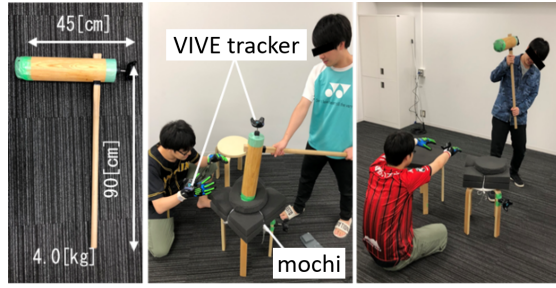


Figure 1: Real Environment System Overview

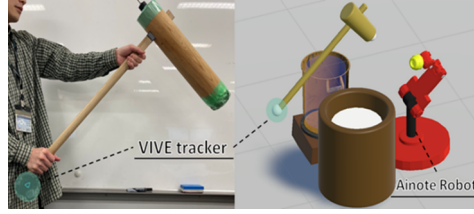


Figure 2: Virtual Reality Environment System Overview

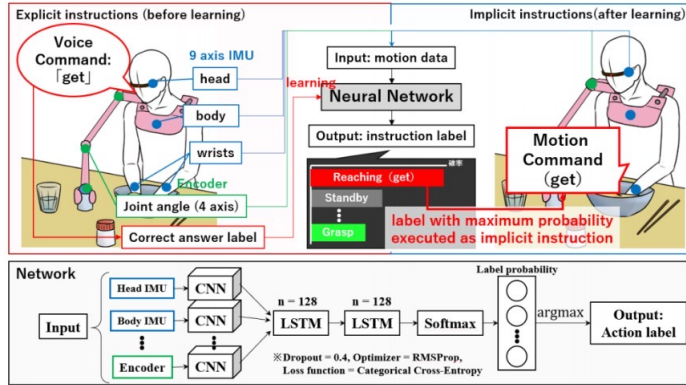


Figure 3: Concept and Network Representation

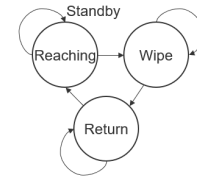
their own rhythm as cautionary measure (in case of any other unexpected changes).

Table 1: Human-Machine Collaboration Qualitative Evaluation Survey

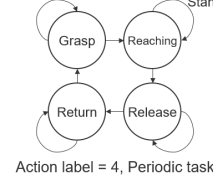
Q1	I was able to predict the action required for the situation
Q2	I was able to modify my behaviour from that of the other party

3.2 Machine understanding ideomotor cues

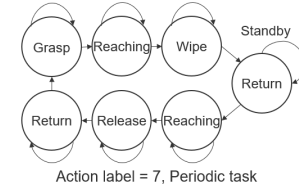
Estimation accuracy of the model (a single model was used for the four different tasks) is displayed on Figure 7. As expected, highest estimation accuracy was obtained on the task with the fewer labels. As the number of labels increased, accuracy decreased. Although Tasks 2 and 4 have the same number of labels (4), as explained above,



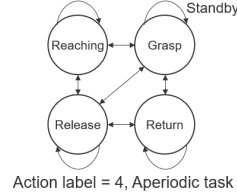
(a) Task 1 - wiping a table (cloth already held)



(b) Task 2 - pick and place



(c) Task 3 - pick cloth up, wipe and put it back



(d) Task 4 - unknown task (decided by the user)

Figure 4: Command Pattern for each Task

the lower F1 score on the last task is due to its aperiodic nature. When paying closer attention to the results, particularly confusion matrices of each task, it was noted that most of the errors in Tasks 1, 2 and 3 were due to the implicit cues being labelled as 0, the label for the "standby". This meant that the primary problem during the collaboration was that the robot arm would sometimes fail to detect the implicit cues (almost certainly related to the amount of available training data). If the cues were detected, however, they were always correctly identified to the appropriate command and therefore followed by the robot behaving according to the user's intention. Despite the occurrence of errors, the performed experiment showed promising results regarding the recognition of implicit cues regardless of the task being executed (no prior information about the task or the context was inputted to the neural network) at a higher level than similar studies. Comparatively, Hayakawa et al. ([5]) used a "Self-organizing Map" (unsupervised learning) to estimate the intention of the operator and have the robot assist in the task. Although the method was designed to be used on a single task (assembly), reported accuracy was of 70%, a lower results than obtained in the present paper for a model designed for estimation on four different tasks.

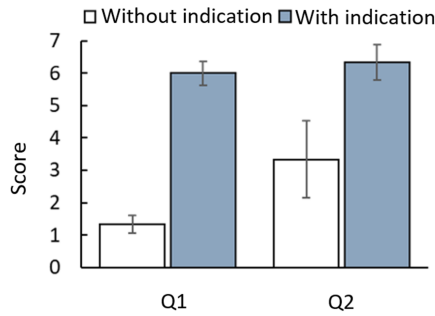


Figure 5: Qualitative Evaluation Answers

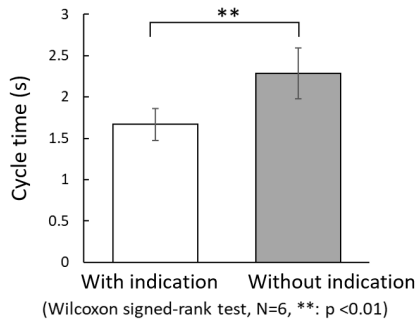


Figure 6: Punch Cycle Response Time

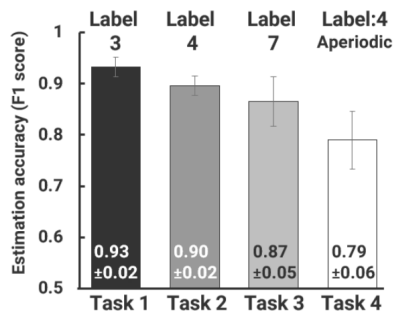


Figure 7: F1 Score on Task Estimation

4 CONCLUSIONS

The key finding of our work is that treating ideomotor reactions a reproducible cues that can be both understood and mimicked by robots has overarching consequences.

One of the goals of artificial intelligence is to use machines as amplifiers of human physical and cognitive capabilities (intelligence amplification). To achieve this, we need to have a sustainable form of human-machine communication. An interaction method that closely adheres to the principles underlying human communication is one with the highest likelihood to succeed. Therefore, we need to consider approaches that allow individuals to communicate with their machines in a similar way that people communicate with each other. More importantly, other existing approaches do not address one of the goals of our work: to create a method that allows the

machine to learn from people without their really trying. In this study, we introduced a prospective body language approach, which allows people to teach artificial agents based on the cues naturally manifested during the process of interaction.

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